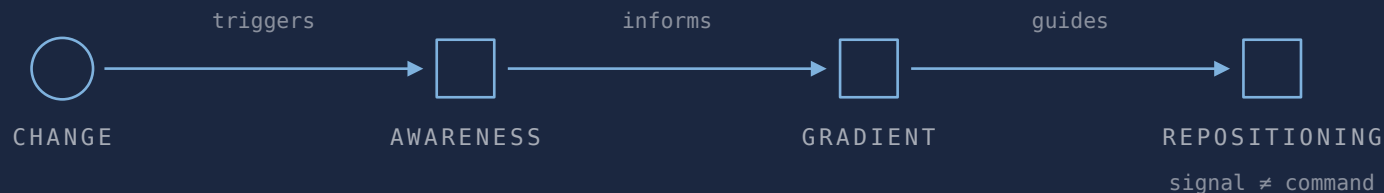


Tegrity.AI

Ecosystem Positioning for Agentic Systems

A proposed architecture for participants that must stay situated, and requalify, as authority, evidence, roles and the ecosystem itself change.



WORKING ARCHITECTURE

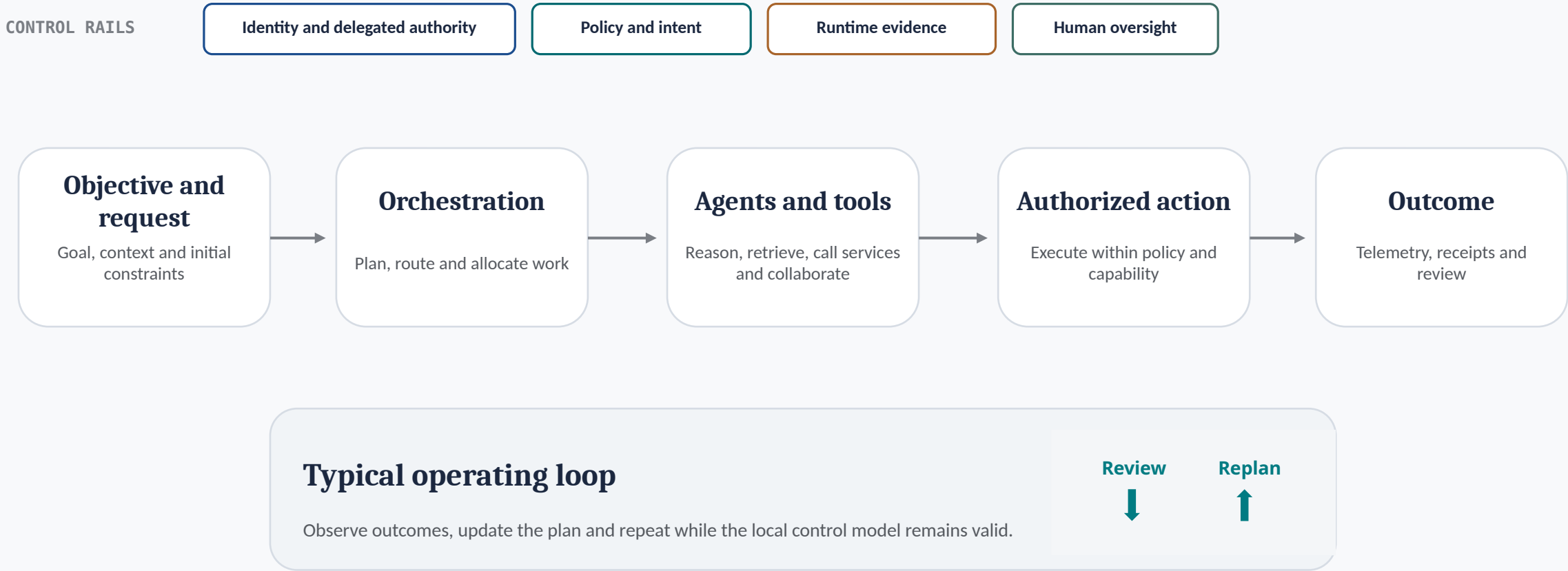
**Proposal stage.
Open for review.**

Ecosystem Awareness, Agentic Citizenship,
Minimum Sufficient Control and Regime Change
Detection coordinated around one participant-local
situational core.

TEGRITY.AI / ECOSYSTEM AWARENESS
SEPTEMBER 2026 · WORKING PROPOSAL

Current Reference Architecture for Agentic AI

A composite of current practice: the strong baseline EA is benchmarked against (arms B1/B2), not a strawman.



The model works while roles, boundaries and the surrounding ecosystem remain legible to the controls that govern the workflow.

Structural problems and current limits

Direct rules and local checks lose explanatory power when participants, roles and dependencies change faster than the control model.

Agent multiplication

More agents, tools and temporary delegates create interaction paths that no operator can review one by one.

Role and identity drift

One participant can hold hybrid roles, change memberships or act under a different authority across interactions.

Ecosystem change

Organizations, dependencies, service availability and external conditions evolve while the system is operating.

Indeterminacy

A local result may be valid but insufficient for the wider decision. Missing scope, provenance or dependency state must remain explicit.

Interpretation gap

Identity, policy and attestation can remain technically valid while their practical meaning changes in the active ecosystem.

Limits of AI oversight

Automated review often inherits the same observation window. Human review adds authority and judgment, but attention, time and context remain finite.

STRUCTURAL CONSEQUENCE

Local correctness and local compliance do not by themselves establish that the composed system is situated correctly for the current decision.

Failure progression: mechanisms and failure types

The ecosystem can drift while local controls keep returning valid-looking answers. The key question is whether uncertainty remains visible or is converted into unsupported closure.

INITIAL STATE

Legible roles and boundaries

Identity and authority are current
Policies match the operating context
Oversight can follow the material paths

DRIFT

The ecosystem changes

Agents multiply and change roles
Boundaries and memberships shift
Signals inherit hidden dependencies
Rules and reviews lag behind

FAILURE

Local closure, systemic indeterminacy

A local PASS travels beyond its scope
Uncertainty and lineage disappear
AI review can repeat the same frame
Human attention saturates

CONDITION AND FAILURE TYPES

TYPE 0 · Structural non-determination

Residual indeterminacy is not removable inside the current representation / capability boundary. More processing of the same evidence cannot manufacture the missing distinction.

Response: preserve the residual, narrow the claim, add a genuinely new information channel or redesign the boundary.

TYPE 1 · Unbounded unresolved uncertainty

Uncertainty is recognized, but no legitimate bounded closure is reached: the system keeps waiting, searching or escalating without an escape condition (indefinite HOLD, paralysis). HOLD is not a safe terminal state.

Response: establish bounded closure: a stop condition plus a qualified posture that keeps the residual explicit.

TYPE 2 · False certainty

The system closes by discarding what it does not know: missing input treated as present, low confidence promoted to certainty, a local result treated as globally valid (forced PASS/FAIL, false closure).

Response: restore uncertainty, scope and provenance, then re-enter the affected scope.

Illustrative failure scenarios

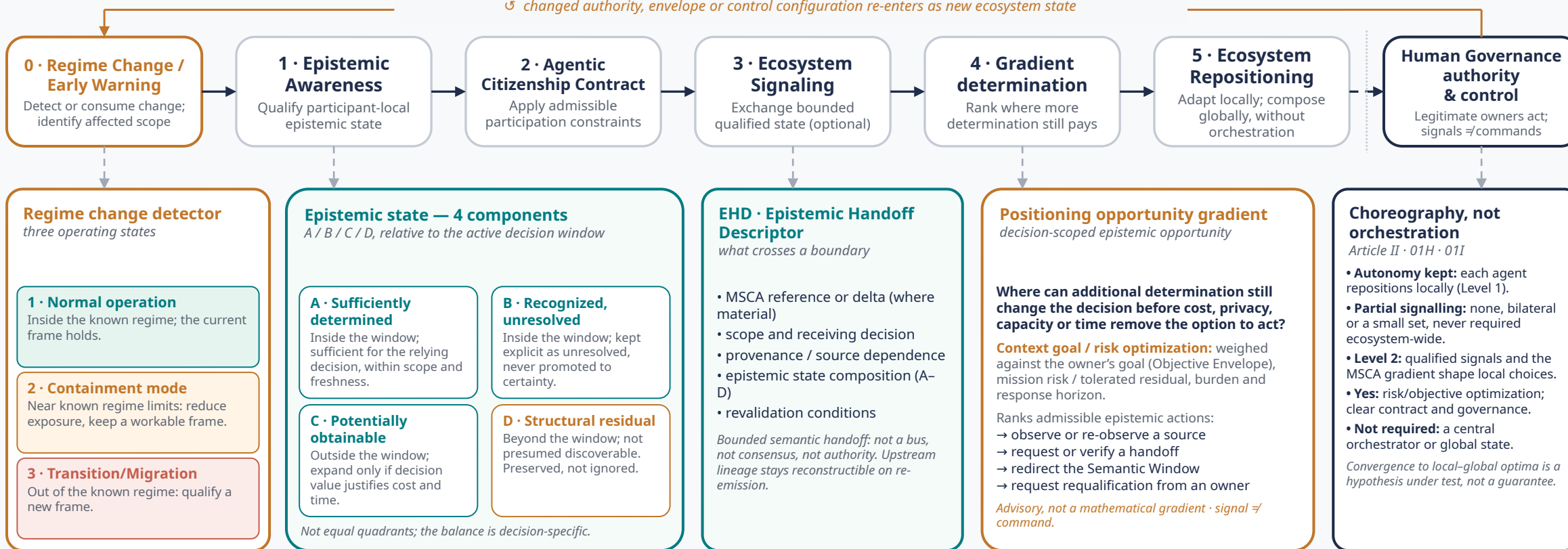
100 Million Tokens — Enterprise evidence compression and false closure under scale. [GitHub reference](#)

Smart-City Mobility Chaos — Distributed mobility drift and cross-actor systemic divergence. [GitHub reference](#)

From unresolved ecosystem change to qualified repositioning

Working proposal, not an adopted standard. Top row: the process sequence. Panels below: expansion of selected phases.

↻ changed authority, envelope or control configuration re-enters as new ecosystem state

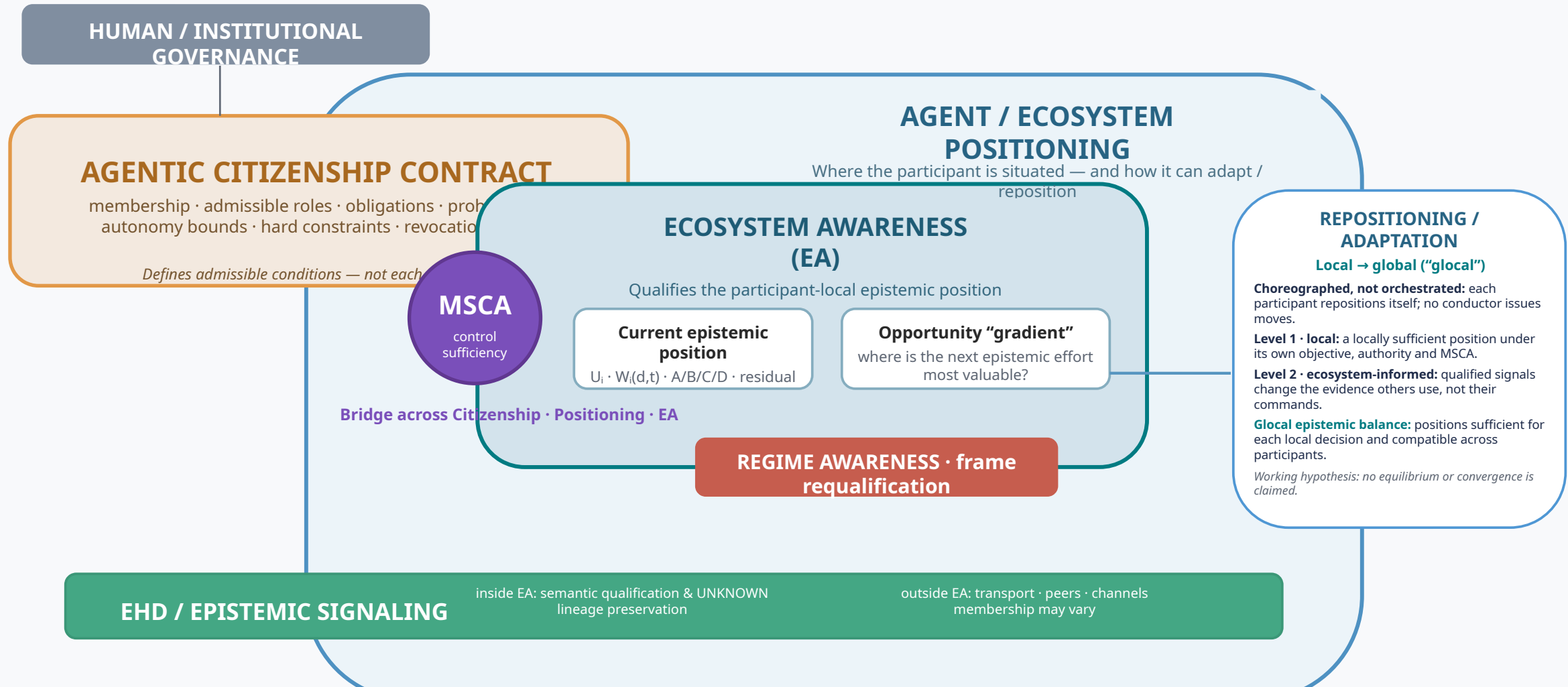


MSCA · Minimum Sufficient Control Architecture

Cross-cutting control-sufficiency check — S Objective Envelope · E operating environment · C coordination scope · P intervention mechanisms · M enabling means
Re-assessed whenever the frame, envelope or authority is requalified. May change observation or coordination means; never creates authority and does not own the Semantic Window.

Architecture map — where each mechanism lives - boundaries

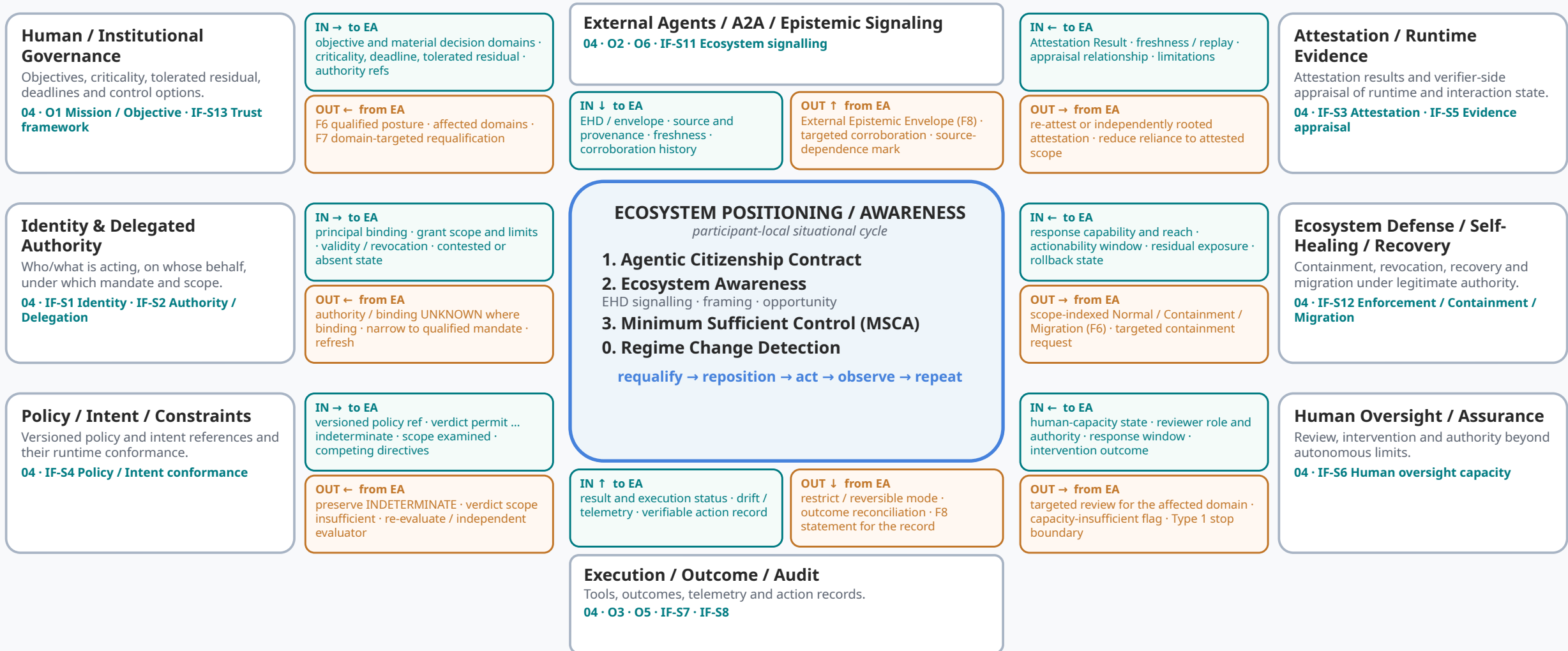
Citizenship defines admissible space; Positioning contains the participant's situated state; EA qualifies what is known and where epistemic effort has value.



Key boundary: Citizenship constrains admissible participation. EA qualifies situated knowledge and opportunity. Positioning includes the actual repositioning. MSCA bridges requirements, knowledge and controllability. EHD crosses boundaries without requiring shared citizenship or a global state.

Ecosystem Positioning / Awareness — subsystem inputs and outputs

Interfaces as defined in corpus doc 04. Green = inputs to EA; orange = EA outputs to that function — requests or qualified state, never commands or authority.

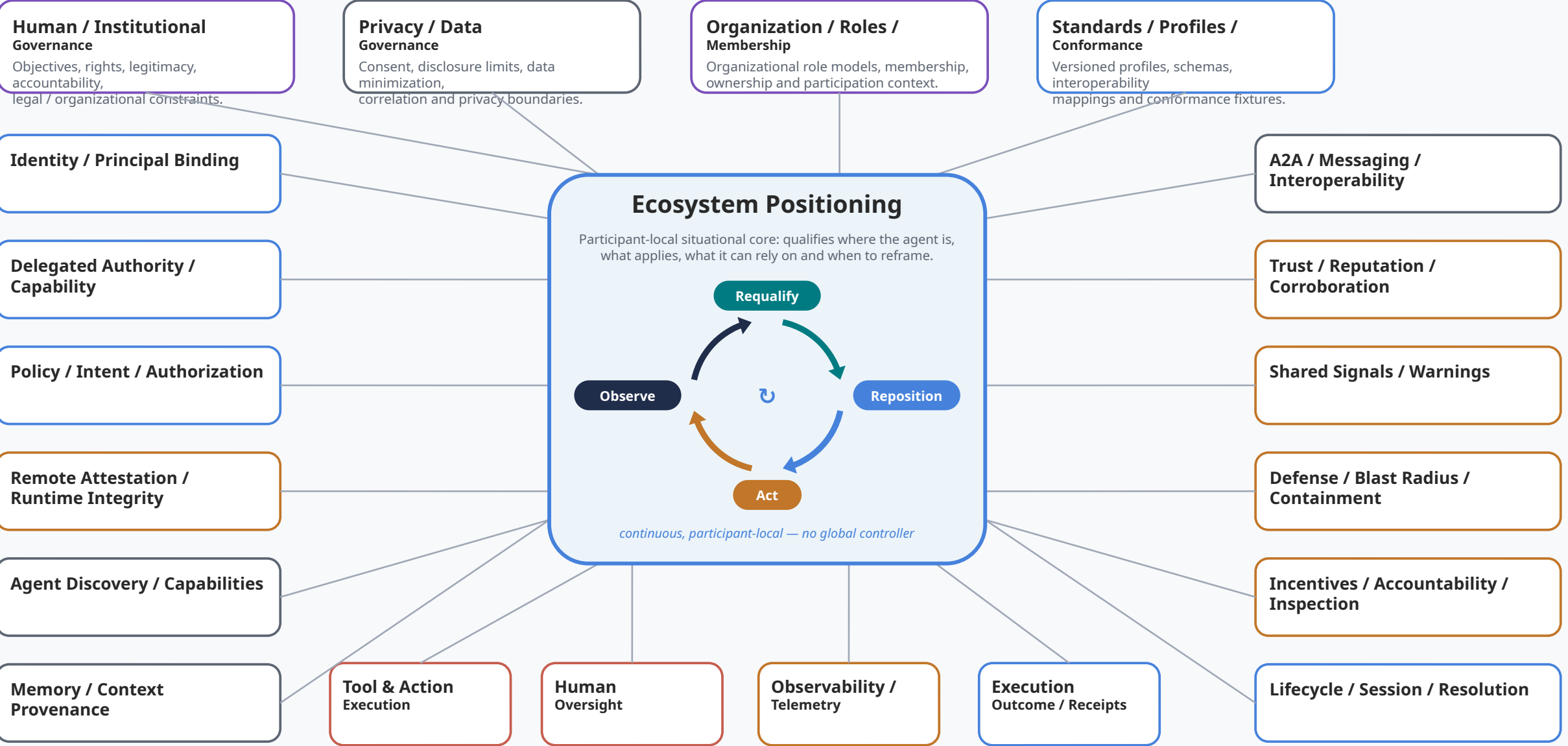


IN · to EA inputs listed in doc 04 for that interface **OUT · from EA** EA outputs to that function **04 · codes** interface IDs O1–O6 / IF-S1–S13

Not shown: O4 context / memory / retrieval, IF-S9 population-level evaluation, IF-S10 privacy and minimum disclosure. F6/F7/F8 = EA functions (posture, requalification, epistemic statement).

Ecosystem Positioning Architecture — Extended Situational Core

Agentic-AI, decentralized-identity (DIF) and ecosystem-defense components around one core: Ecosystem Positioning.



Reading rule: outer boxes are familiar agentic-AI capabilities. Positioning composes their decision-relevant state into a participant-local frame and requests requalification; owners decide.

TIDA Component: Ecosystem Awareness

A candidate system-level epistemic architecture for independently governed human and agentic components.

SCOPED EPISTEMIC STATE What is determined, unresolved, potentially knowable and structurally residual.	NON-FUNGIBLE COMPOSITION Confidence in one domain cannot erase uncertainty in another.	RISK / CAPACITY REQUALIFICATION Expand, narrow or stop determination according to mission risk and response horizon.	TRANSPORT-NEUTRAL HANDOFF Preserve the minimum semantics needed across organisational and technical boundaries.
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ARCHITECTURAL ALIGNMENT

FG-TIDA

Public exchanges and cross-theme mappings indicate FG-TIDA-wide architectural alignment in progress; formal adoption is not claimed.

Tegrity.AI · AI Integrity Management · Public References

Public research, architecture and pre-standardization routes for Ecosystem Awareness, Minimum Sufficient Control Architecture and Regime Awareness.

Operational Integrity Management

It connects three coordinated components:

- 01 · Ecosystem Awareness
- 02 · Minimum Sufficient Control Architecture (MSCA)
- 03 · Regime Awareness / Sufficiently Good Early Warning

Agentic Citizenship (annex 01I) is treated as a separate participation layer.

Working architecture for review, benchmark construction and testing. Public work does not imply standards-body adoption or endorsement.

[Structural Awareness Programme — parent route](#)

[Return to the four-part Structural Awareness programme and its human-readable structure.](#)

[Ecosystem Awareness — public corpus](#)

[Decision-scoped epistemic qualification and the maintained EA reading route.](#)

[MSCA — public corpus](#)

[Minimum Sufficient Control Architecture public reading route.](#)

[Regime Awareness / Regime Change — public route](#)

[Regime Awareness / Sufficiently Good Early Warning reading route.](#)

[Canonical Architecture Benchmark & Evidence v0.2](#)

B0–B3 comparator contract, hypotheses, falsifiers and evidence grades.

[FG-TIDA themes repository](#)

Public theme discussions and cross-theme interfaces.